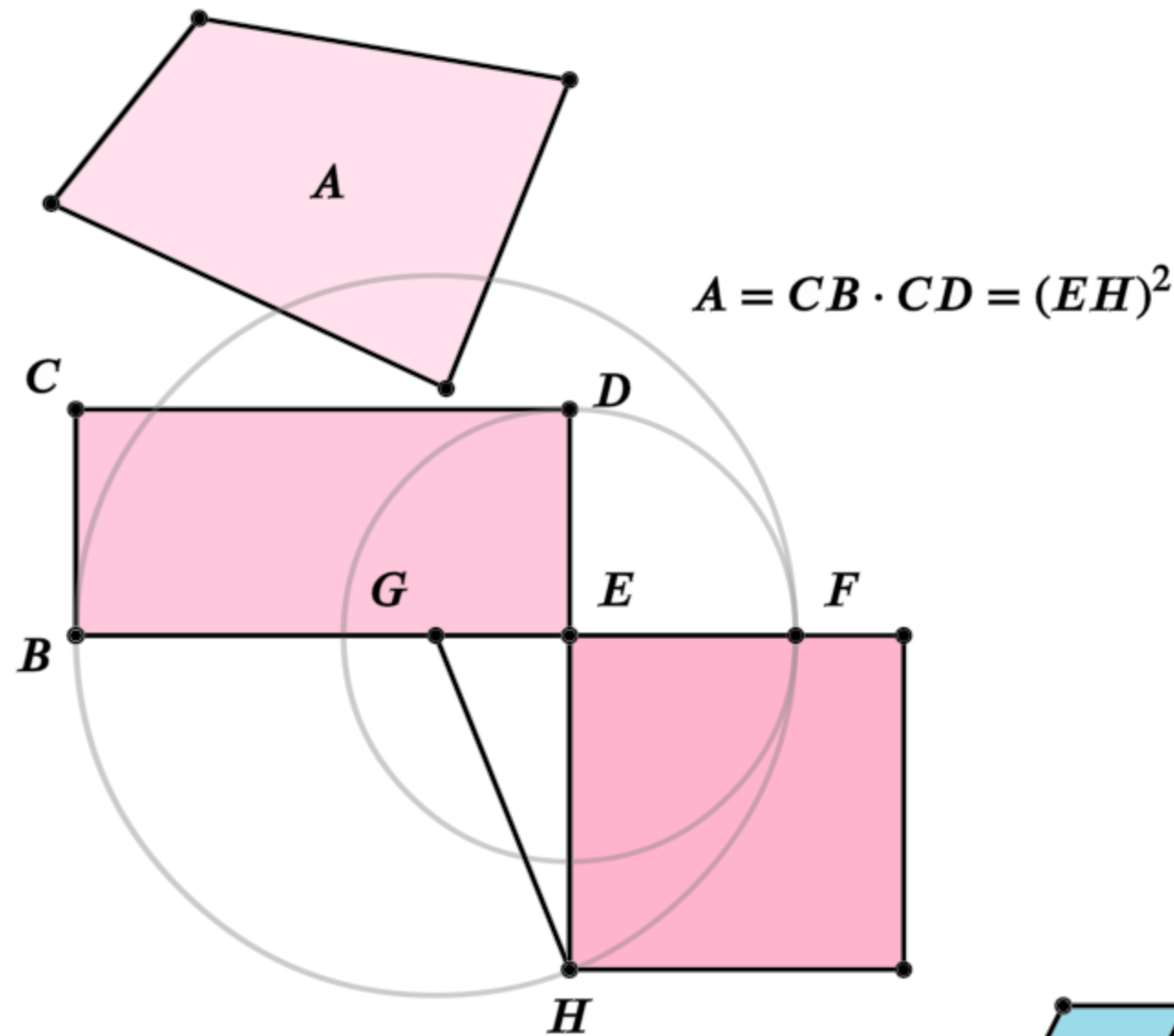


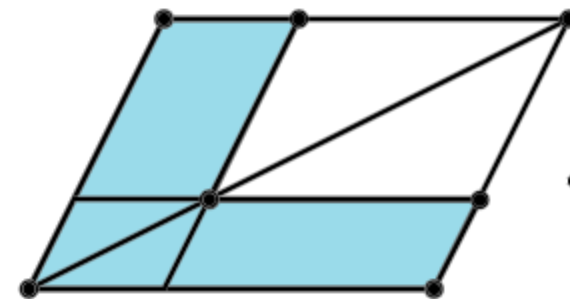
Euclid's Elements

Book II



It is a remarkable fact in the history of geometry, that the Elements of Euclid, written two thousand years ago, are still regarded by many as the best introduction to the mathematical sciences.

- Florian Cajori,
A History of Mathematics (1893)

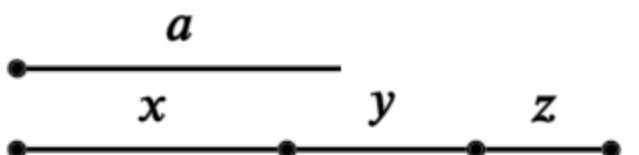


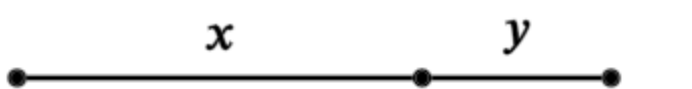
Definitions:

Any rectangular parallelogram is said to be contained by the two straight lines containing the right angle.

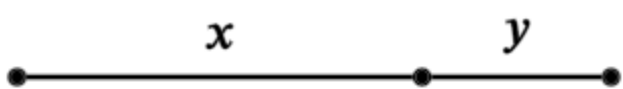
And in any parallelogrammic area let any one whatever of the parallelograms about its diameter with the two complements be called a gnomon.

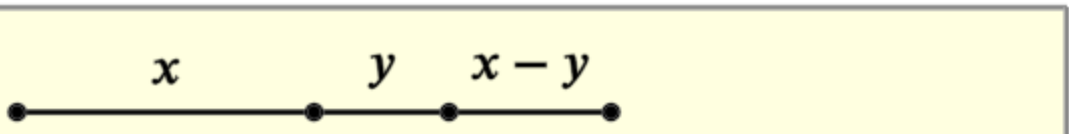
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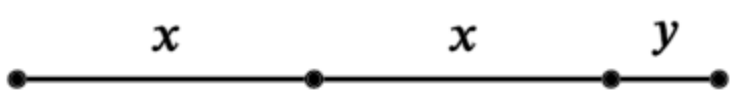
1. 
 $a(x + y + z) = ax + ay + az$

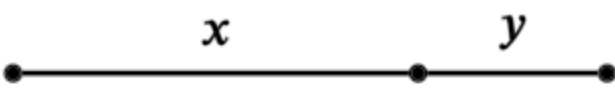
2. 
 $(x + y)^2 = x(x + y) + y(x + y)$

3. 
 $y(x + y) = yx + y^2$

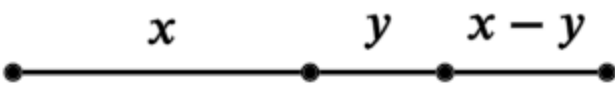
4. 
 $(x + y)^2 = x^2 + y^2 + 2xy$

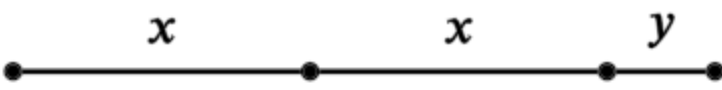
5. 
 $(x + y)(x - y) + y^2 = x^2$

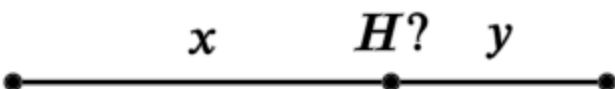
6. 
 $y(2x + y) + x^2 = (x + y)^2$

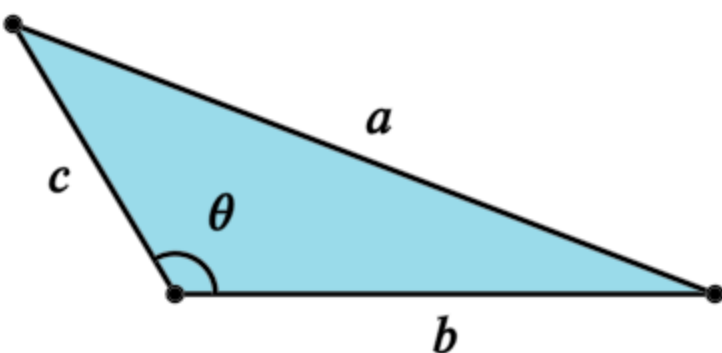
7. 
 $(x + y)^2 + y^2 = x^2 + 2y(x + y)$

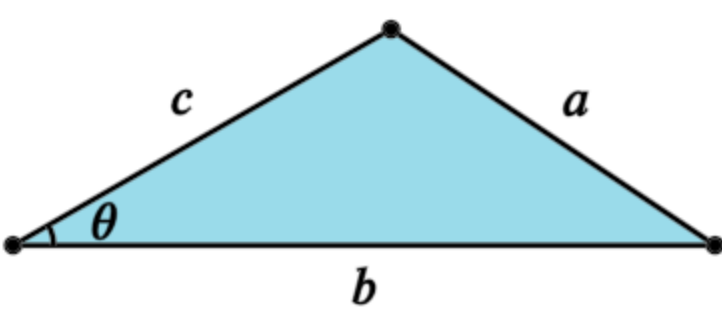
8. 
 $(x + 2y)^2 = 4y(x + y) + x^2$

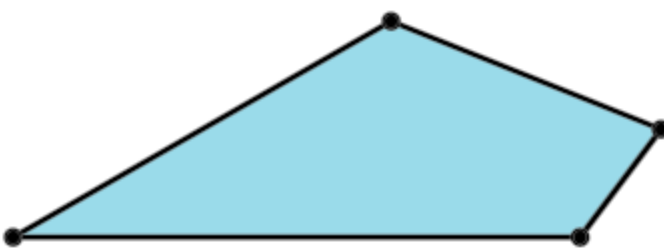
9. 
 $(x + y)^2 + (x - y)^2 = 2(x^2 + y^2)$

10. 
 $(2x + y)^2 + y^2 = 2(x^2 + (x + y)^2)$

11. 
Find H such that: $y(x + y) = x^2$

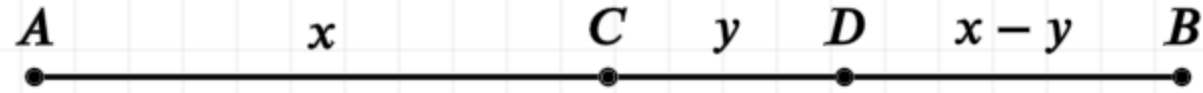
12. 
Cosine Law, $a^2 = b^2 + c^2 - 2bc \cos \theta$

13. 
Cosine Law. $a^2 = b^2 + c^2 - 2bc \cos \theta$

14. 
Construct a square equal to the polygon

Proposition 5 of Book 2

If a straight line be cut into equal and unequal segments, the rectangle contained by the unequal segments of the whole together with the square on the straight line between the points of section is equal to the square on the half



In other words:

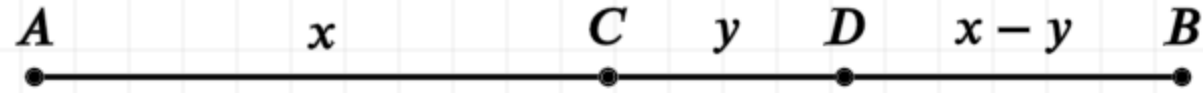
Let AB be a straight line, bisected at point C, and cut at an arbitrary point D

$$AC = CB, \quad AD = AC + CD, \quad DB = CB - CD$$

$$AC = x, \quad CB = x, \quad CD = y$$

Proposition 5 of Book 2

If a straight line be cut into equal and unequal segments, the rectangle contained by the unequal segments of the whole together with the square on the straight line between the points of section is equal to the square on the half



In other words:

Let AB be a straight line, bisected at point C, and cut at an arbitrary point D

The rectangle formed by the uneven segments (AD and DB) added to the square of the tiny segment CD, is equal to the half segment (CB) all squared.

$$AC = CB, \quad AD = AC + CD, \quad DB = CB - CD$$

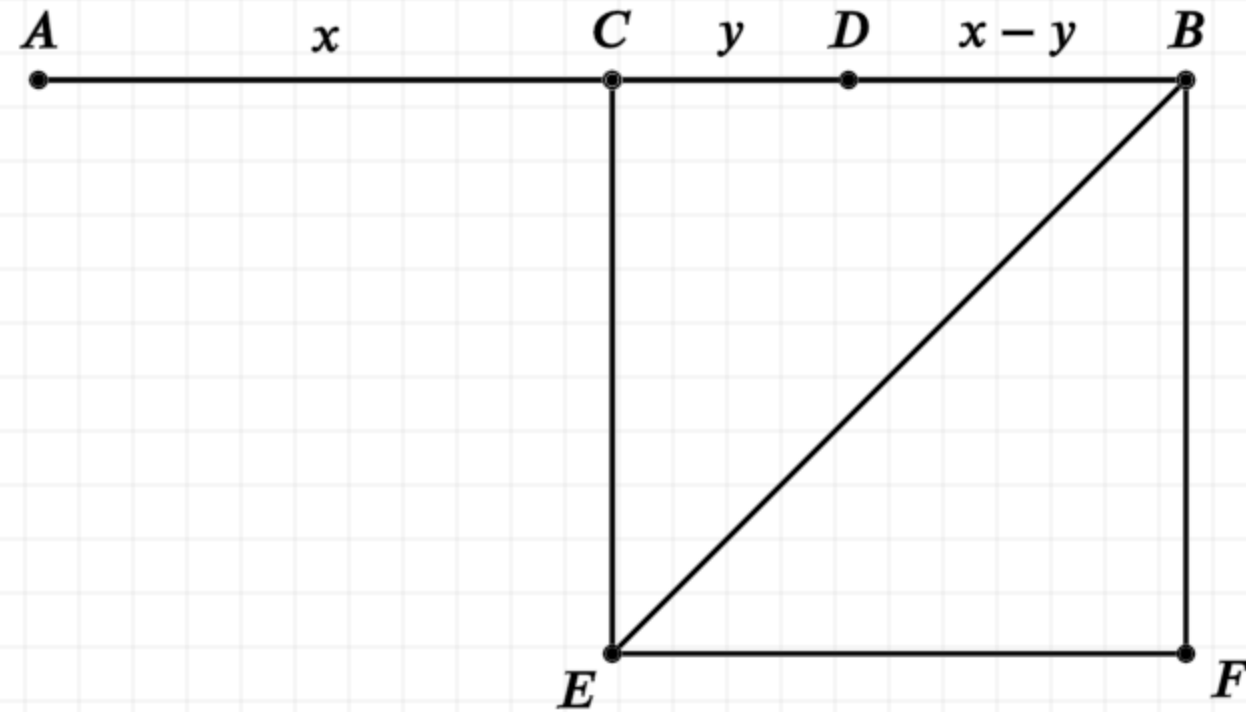
$$AD \cdot DB + CD^2 = CB^2$$

$$AC = x, \quad CB = x, \quad CD = y$$

$$(x + y) \cdot (x - y) + y^2 = x^2$$

Proposition 5 of Book 2

If a straight line be cut into equal and unequal segments, the rectangle contained by the unequal segments of the whole together with the square on the straight line between the points of section is equal to the square on the half



$$AC = CB, \quad AD = AC + CD, \quad DB = CB - CD$$

In other words:

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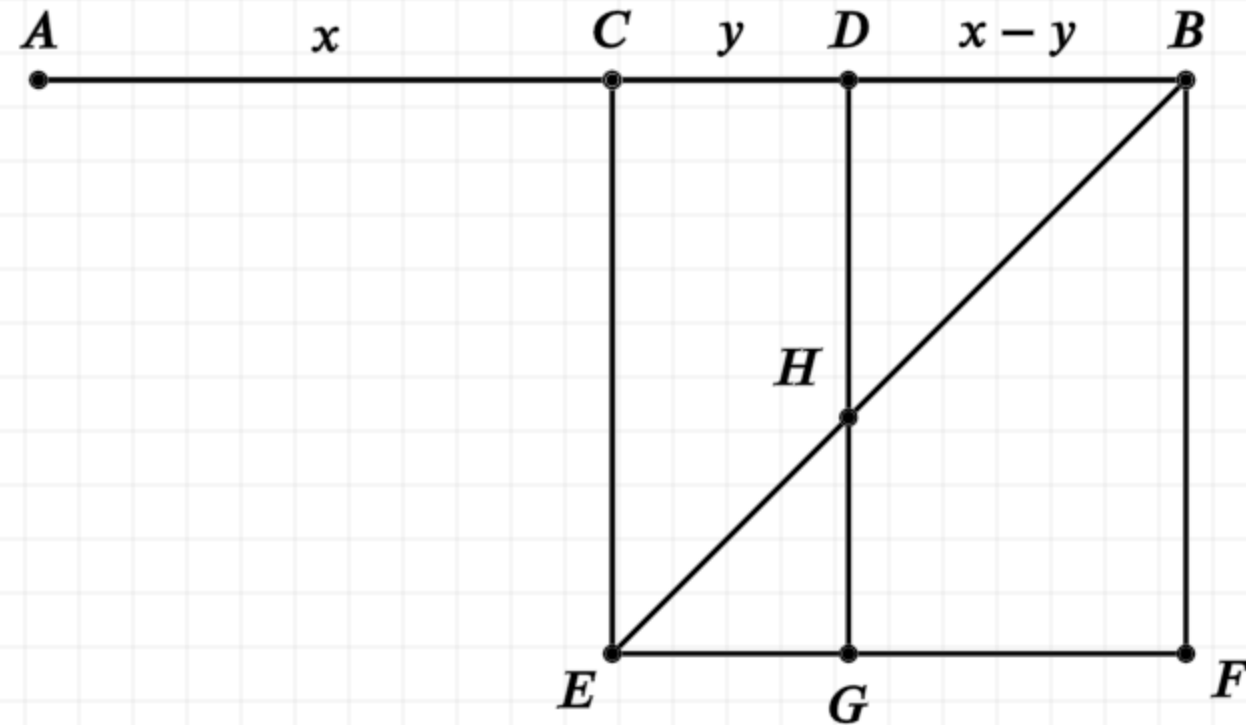
$$AD \cdot DB + CD^2 = CB^2$$

Construction:

Draw a square CEFB on the line CB (I.46) and draw the diagonal BE

Proposition 5 of Book 2

If a straight line be cut into equal and unequal segments, the rectangle contained by the unequal segments of the whole together with the square on the straight line between the points of section is equal to the square on the half



$$AC = CB, \quad AD = AC + CD, \quad DB = CB - CD$$

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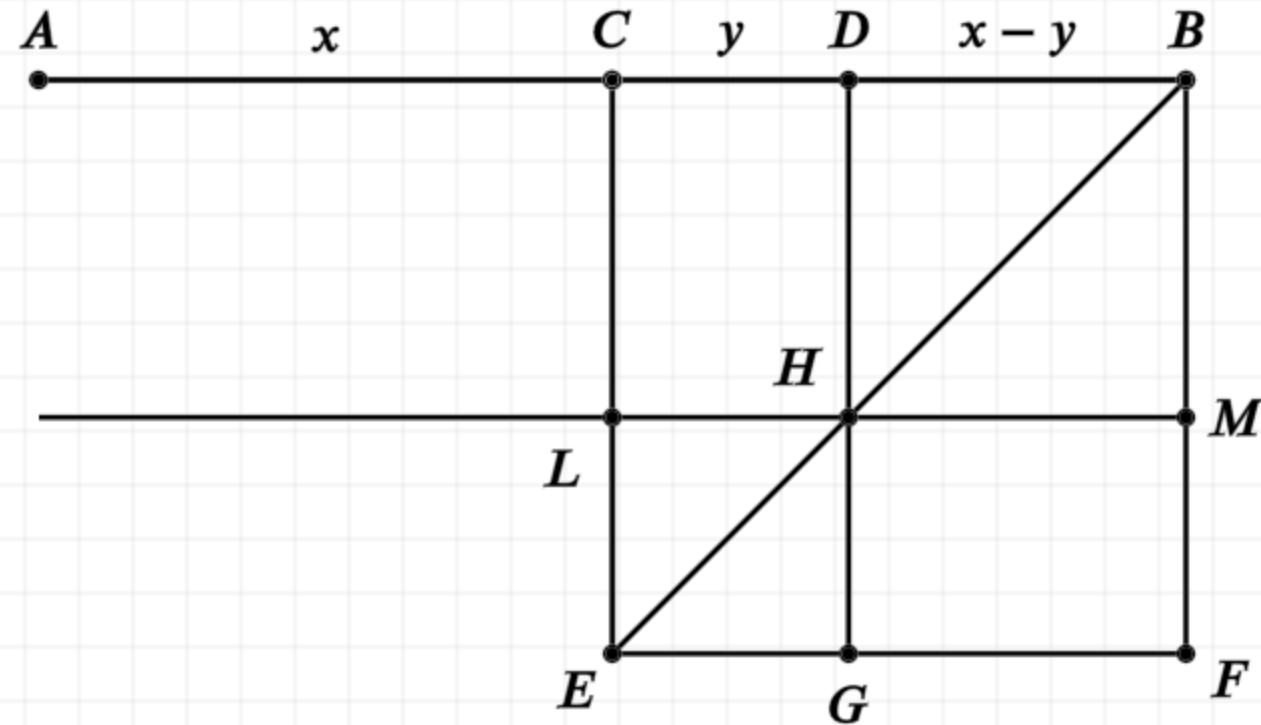
Construction:

Draw a square CEFB on the line CB (I.46) and draw the diagonal BE

From point D, draw a line parallel to either CE or BF (I.31)

Proposition 5 of Book 2

If a straight line be cut into equal and unequal segments, the rectangle contained by the unequal segments of the whole together with the square on the straight line between the points of section is equal to the square on the half



$$AC = CB, \quad AD = AC + CD, \quad DB = CB - CD$$

In other words:

Let AB be a straight line, bisected at point C, and cut at an arbitrary point D

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Construction:

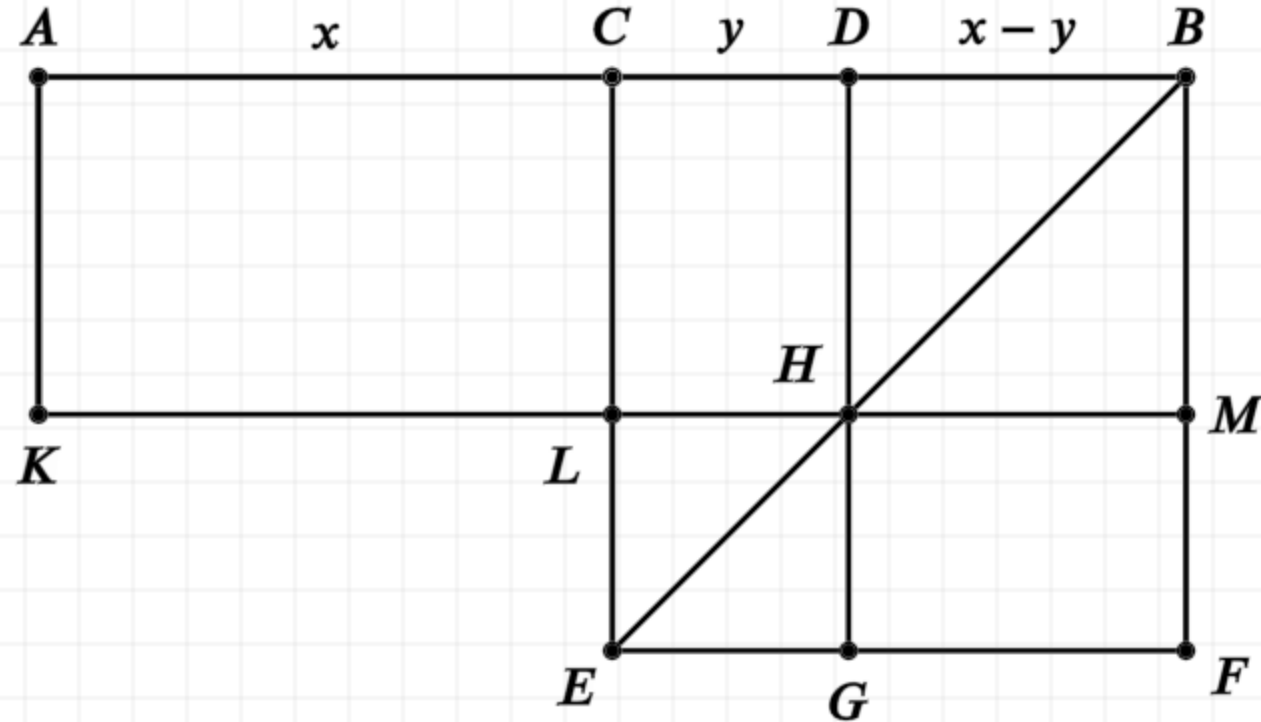
Draw a square CEFB on the line CB (I.46) and draw the diagonal BE

From point D, draw a line parallel to either CE or BF (I.31)

From point H, draw a line parallel to either AB or EF (I.31)

Proposition 5 of Book 2

If a straight line be cut into equal and unequal segments, the rectangle contained by the unequal segments of the whole together with the square on the straight line between the points of section is equal to the square on the half



$$AC = CB, \quad AD = AC + CD, \quad DB = CB - CD$$

In other words:

Let AB be a straight line, bisected at point C, and cut at an arbitrary point D

$$AD \cdot DB + CD^2 = CB^2$$

Construction:

Draw a square CEFB on the line CB (I.46) and draw the diagonal BE

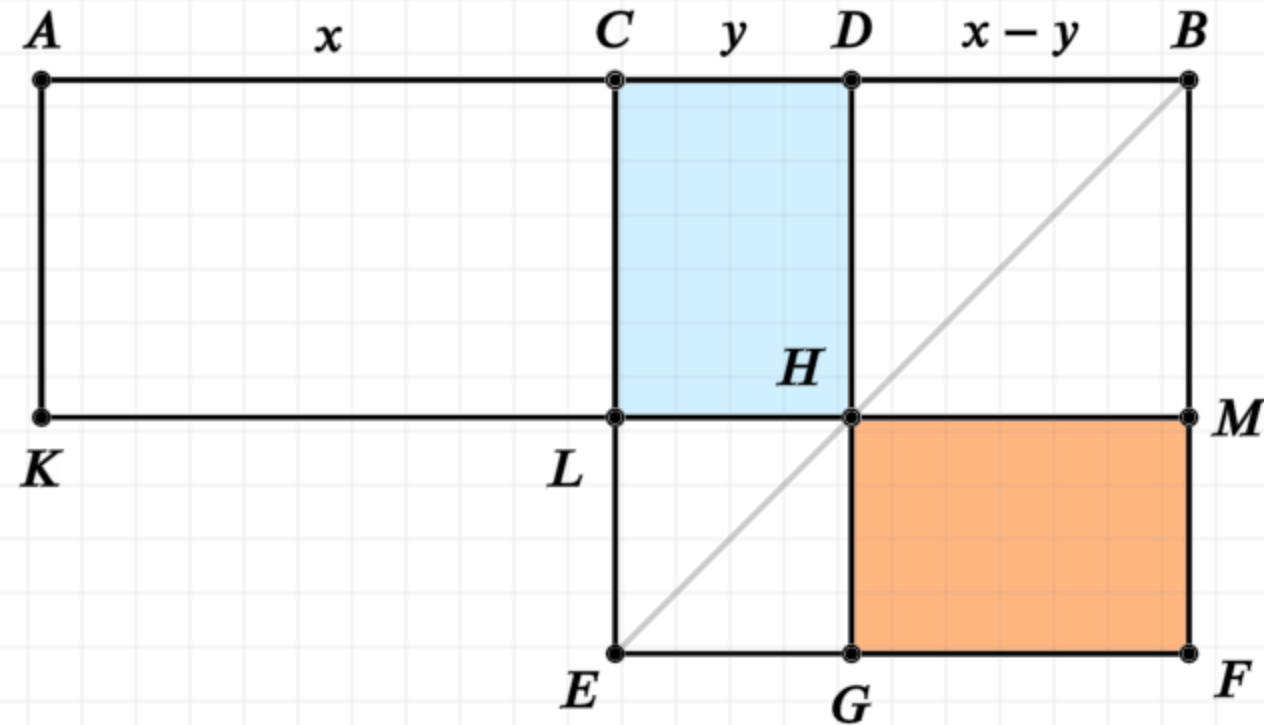
From point D, draw a line parallel to either CE or BF (I.31)

From point H, draw a line parallel to either AB or EF (I.31)

From point A, draw a line parallel to either CL or BM (I.31)

Proposition 5 of Book 2

If a straight line be cut into equal and unequal segments, the rectangle contained by the unequal segments of the whole together with the square on the straight line between the points of section is equal to the square on the half



$$AC = CB, \quad AD = AC + CD, \quad DB = CB - CD$$

$$\text{CH} = \text{HF}$$

In other words:

Let AB be a straight line, bisected at point C, and cut at an arbitrary point D

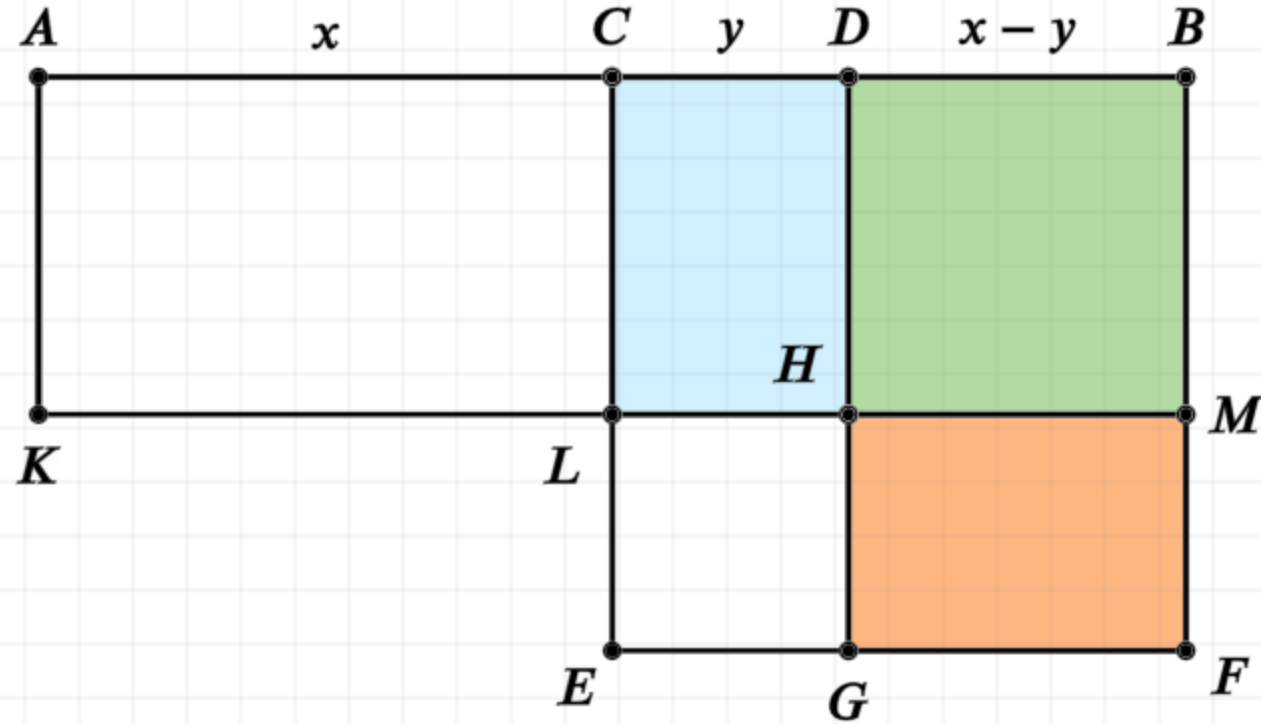
$$AD \cdot DB + CD^2 = CB^2$$

Proof:

The complements CH and HF are equal (I.43), and if we add the rectangle DM, then the rectangles CM and DF are equal

Proposition 5 of Book 2

If a straight line be cut into equal and unequal segments, the rectangle contained by the unequal segments of the whole together with the square on the straight line between the points of section is equal to the square on the half



$$AC = CB, \quad AD = AC + CD, \quad DB = CB - CD$$

$$\blacksquare CH = \blacksquare HF$$

$$CH + DM = HF + DM$$

In other words:

Let AB be a straight line, bisected at point C, and cut at an arbitrary point D

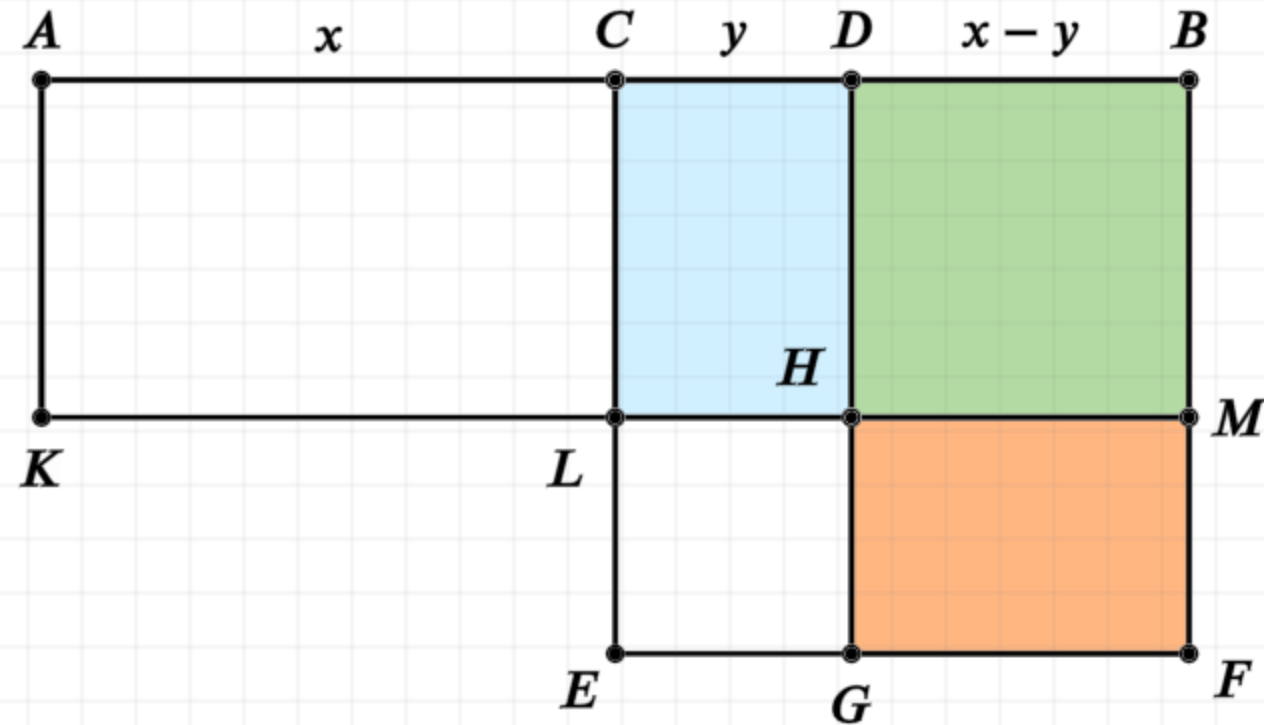
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$$AC = CB, \quad AD = AC + CD, \quad DB = CB - CD$$

$$\text{CH} = \text{HF}$$

$$\text{CH} + \text{DM} = \text{HF} + \text{DM} : \quad \therefore \text{CM} = \text{DF}$$

In other words:

Let AB be a straight line, bisected at point C, and cut at an arbitrary point D

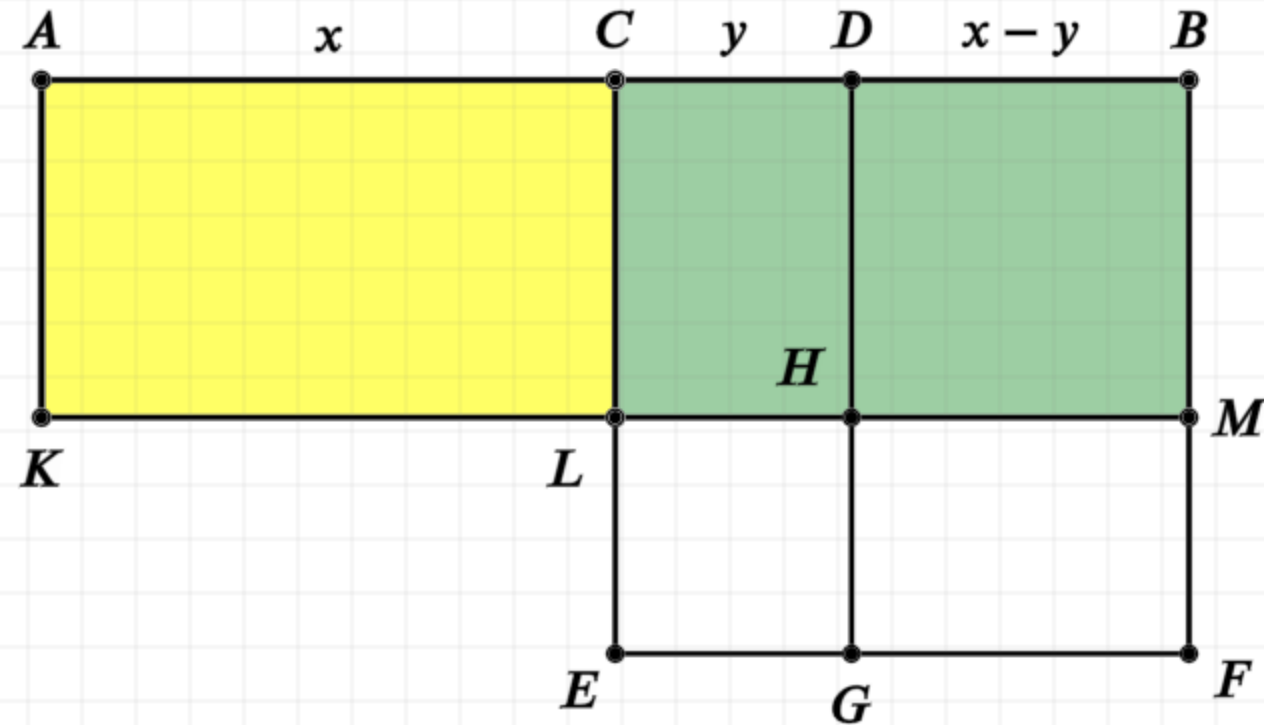
$$AD \cdot DB + CD^2 = CB^2$$

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Proposition 5 of Book 2

If a straight line be cut into equal and unequal segments, the rectangle contained by the unequal segments of the whole together with the square on the straight line between the points of section is equal to the square on the half



$$AC = CB, \quad AD = AC + CD, \quad DB = CB - CD$$

$$CH = HF$$

$$CH + DM = HF + DM : \quad \therefore CM = DF$$

$$AL = CM$$

In other words:

Let AB be a straight line, bisected at point C, and cut at an arbitrary point D

$$AD \cdot DB + CD^2 = CB^2$$

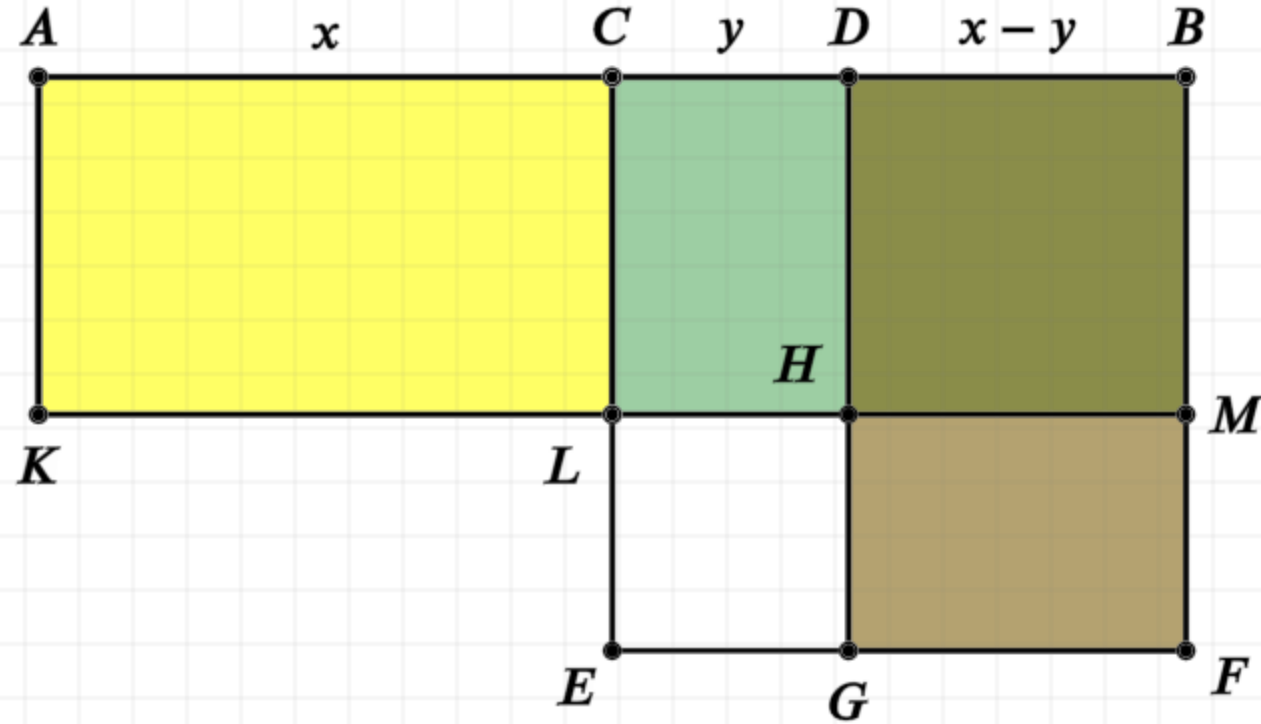
Proof:

The complements CH and HF are equal (I.43), and if we add the rectangle DM, then the rectangles CM and DF are equal

The rectangles CM and AL are equal (I.36)

Proposition 5 of Book 2

If a straight line be cut into equal and unequal segments, the rectangle contained by the unequal segments of the whole together with the square on the straight line between the points of section is equal to the square on the half


$$AC = CB, \quad AD = AC + CD, \quad DB = CB - CD$$
$$\text{CH} = \text{HF}$$
$$CH + \text{DM} = HF + \text{DM} : \therefore \text{CM} = \text{DF}$$

AL = CM = DF

In other words:

Let AB be a straight line, bisected at point C, and cut at an arbitrary point D

$$AD \cdot DB + CD^2 = CB^2$$

Proof:

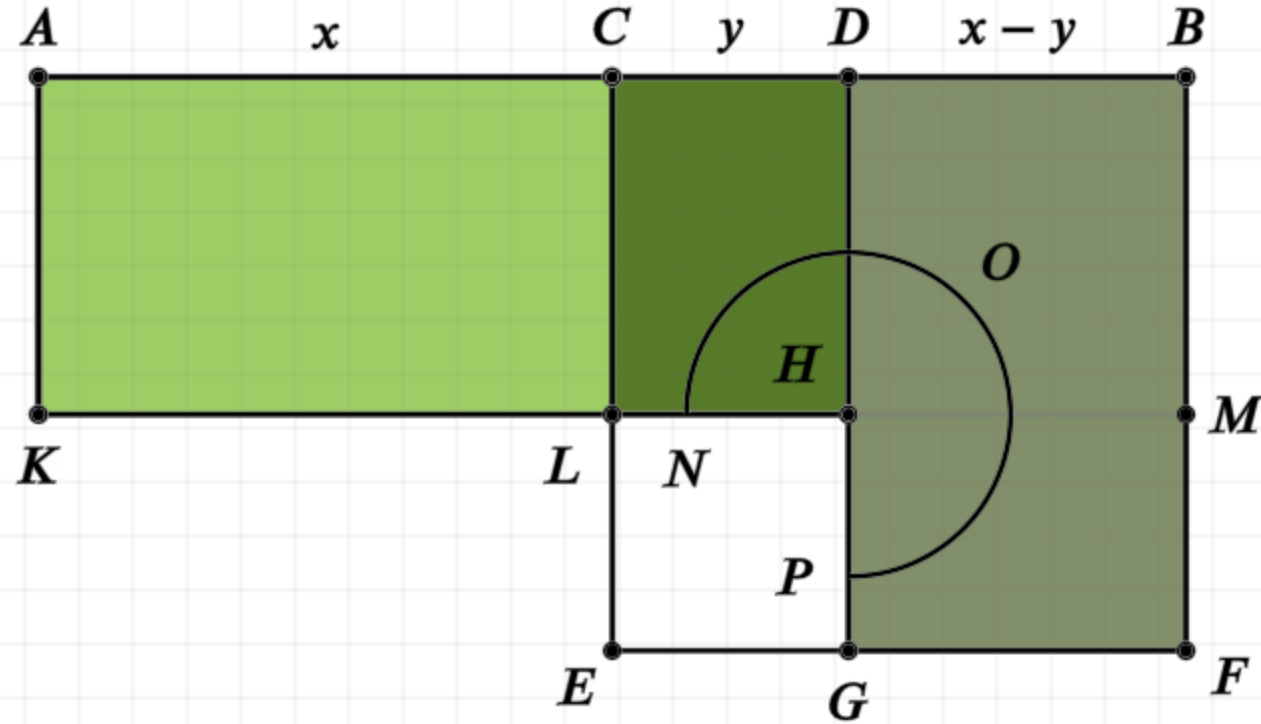
The complements CH and HF are equal (I.43), and if we add the rectangle DM, then the rectangles CM and DF are equal

The rectangles CM and AL are equal (I.36)

which means that AL and DF are also equal

Proposition 5 of Book 2

If a straight line be cut into equal and unequal segments, the rectangle contained by the unequal segments of the whole together with the square on the straight line between the points of section is equal to the square on the half



$$AC = CB, \quad AD = AC + CD, \quad DB = CB - CD$$

$$CH = HF$$

$$CH + DM = HF + DM : \quad \therefore CM = DF$$

$$AL = CM = DF$$

$$AL + CH = DF + CH$$

$$AH = NOP$$

In other words:

Let AB be a straight line, bisected at point C, and cut at an arbitrary point D

$$AD \cdot DB + CD^2 = CB^2$$

Proof:

The complements CH and HF are equal (I.43), and if we add the rectangle DM, then the rectangles CM and DF are equal

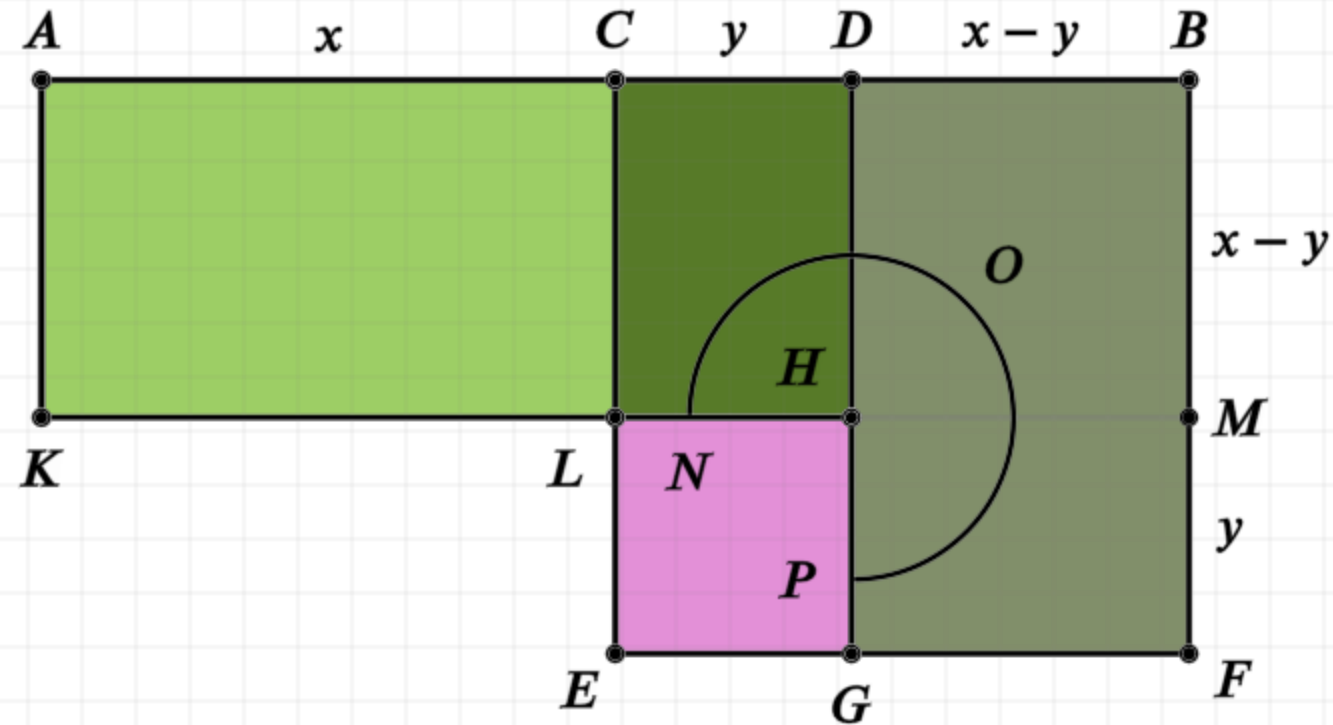
The rectangles CM and AL are equal (I.36)

which means that AL and DF are also equal

Let CH be added to each of AL and DF. Now AH is equal to gnomon NOP

Proposition 5 of Book 2

If a straight line be cut into equal and unequal segments, the rectangle contained by the unequal segments of the whole together with the square on the straight line between the points of section is equal to the square on the half



$$AC = CB, \quad AD = AC + CD, \quad DB = CB - CD$$

$$CH = HF$$

$$CH + \text{DM} = HF + \text{DM} : \therefore CM = DF$$

$AL = CM = DF$

$$AL + CH = DF + CH$$

$AH = NOP$

$$DH = DB, \quad LG = CD^2$$

In other words:

Let AB be a straight line, bisected at point C, and cut at an arbitrary point D

$$AD \cdot DB + CD^2 = CB^2$$

Proof:

The complements CH and HF are equal (I.43), and if we add the rectangle DM, then the rectangles CM and DF are equal

The rectangles CM and AL are equal (I.36)

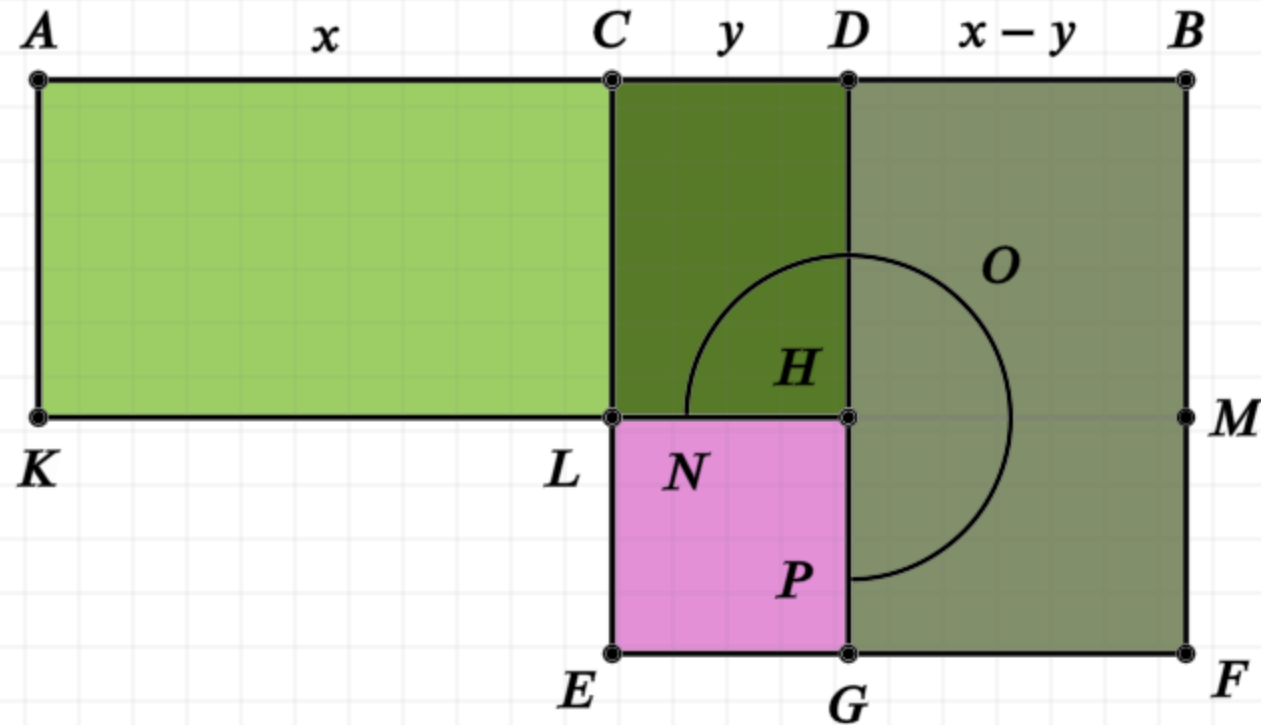
which means that AL and DF are also equal

Let CH be added to each of AL and DF. Now AH is equal to gnomon NOP

For a proof showing that DM and LG are squares, see II.4

Proposition 5 of Book 2

If a straight line be cut into equal and unequal segments, the rectangle contained by the unequal segments of the whole together with the square on the straight line between the points of section is equal to the square on the half



$$AC = CB, \quad AD = AC + CD, \quad DB = CB - CD$$

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$AL = CM = DF$

$$AL + CH = DF + CH$$

■ $AH =$ ■ NOP

$$DH = DB, \quad LG = CD^2$$

$$\blacksquare AH = AD \cdot DB = \blacksquare NOP$$

In other words:

Let AB be a straight line, bisected at point C, and cut at an arbitrary point D

$$AD \cdot DB + CD^2 = CB^2$$

Proof:

The complements CH and HF are equal (I.43), and if we add the rectangle DM, then the rectangles CM and DF are equal

The rectangles CM and AL are equal (I.36)

which means that AL and DF are also equal

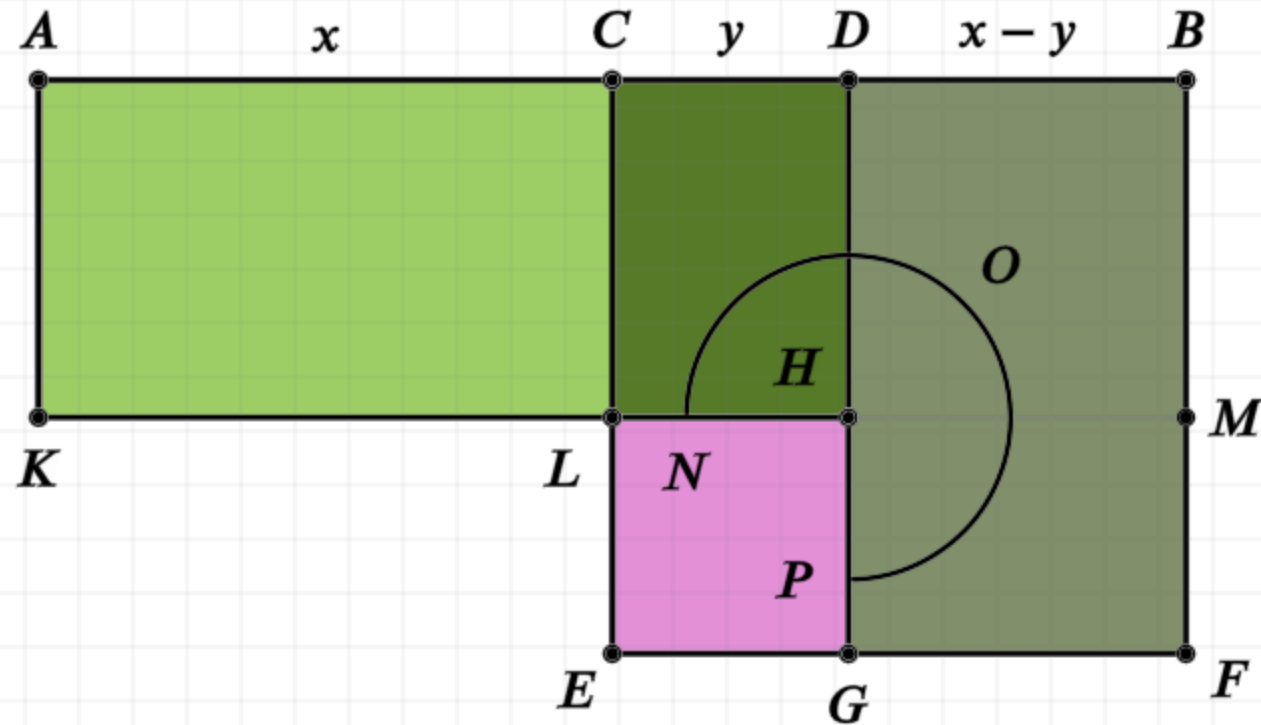
Let CH be added to each of AL and DF. Now AH is equal to gnomon NOP

For a proof showing that DM and LG are squares, see II.4

AH is equal to the rectangle formed by AD, DH , and also by AD, DB , therefore $AD \cdot DB$ is equal to the gnomon NOP

Proposition 5 of Book 2

If a straight line be cut into equal and unequal segments, the rectangle contained by the unequal segments of the whole together with the square on the straight line between the points of section is equal to the square on the half



$$AC = CB, \quad AD = AC + CD, \quad DB = CB - CD$$

$$CH = HF$$

$$CH + DM = HF + DM : \quad \therefore CM = DF$$

$$AL = CM = DF$$

$$AL + CH = DF + CH$$

$$AH = NOP$$

$$DH = DB, \quad LG = CD^2$$

$$AH = AD \cdot DB = NOP$$

$$AD \cdot DB + CD^2 = NOP + LG = CF$$

In other words:

Let AB be a straight line, bisected at point C, and cut at an arbitrary point D

$$AD \cdot DB + CD^2 = CB^2$$

Proof:

The complements CH and HF are equal (I.43), and if we add the rectangle DM, then the rectangles CM and DF are equal

The rectangles CM and AL are equal (I.36)

which means that AL and DF are also equal

Let CH be added to each of AL and DF. Now AH is equal to gnomon NOP

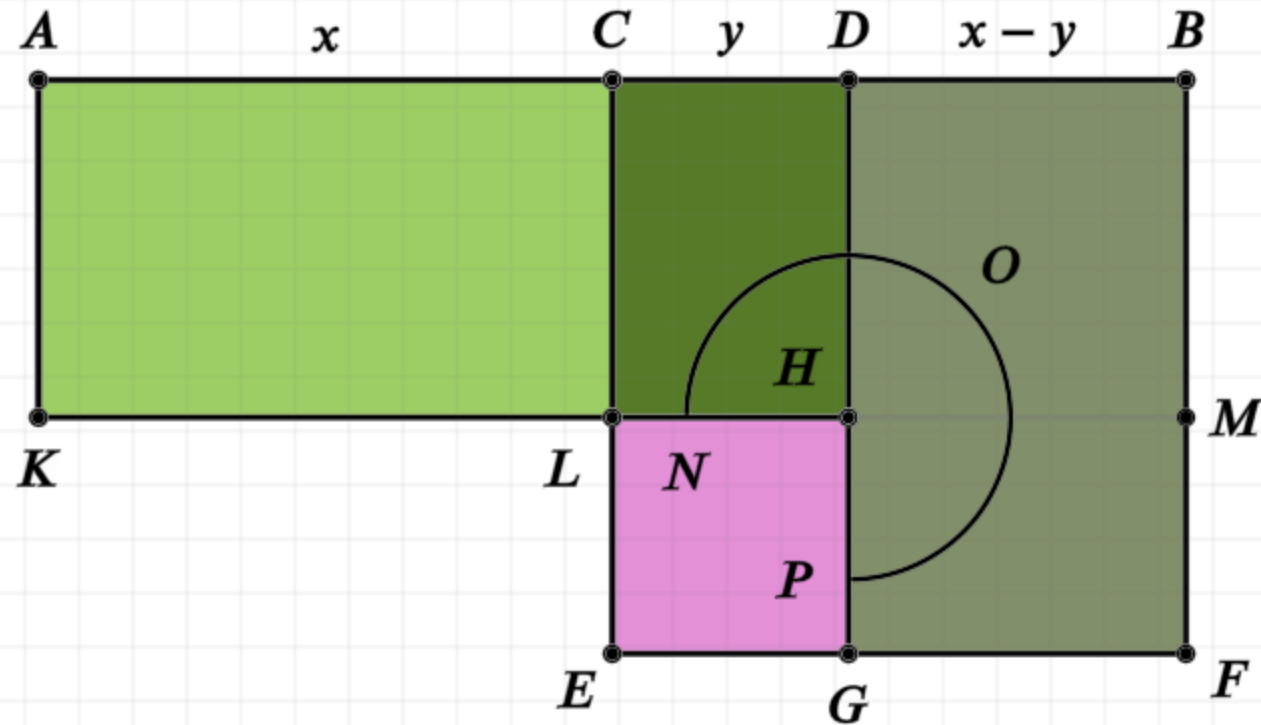
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AH is equal to the rectangle formed by AD, DH, and also by AD, DB, therefore $AD \cdot DB$ is equal to the gnomon NOP

LG is equal to the square on CD, add it to both AH and NOP, retaining the equality

Proposition 5 of Book 2

If a straight line be cut into equal and unequal segments, the rectangle contained by the unequal segments of the whole together with the square on the straight line between the points of section is equal to the square on the half



$$AC = CB, \quad AD = AC + CD, \quad DB = CB - CD$$

$$CH = HF$$

$$CH + DM = HF + DM : \quad \therefore CM = DF$$

$$AL = CM = DF$$

$$AL + CH = DF + CH$$

$$AH = NOP$$

$$DH = DB, \quad LG = CD^2$$

$$AH = AD \cdot DB = NOP$$

$$AD \cdot DB + CD^2 = NOP + LG = CF$$

$$AD \cdot DB + CD^2 = CB^2$$

In other words:

Let AB be a straight line, bisected at point C, and cut at an arbitrary point D

$$AD \cdot DB + CD^2 = CB^2$$

Proof:

The complements CH and HF are equal (I.43), and if we add the rectangle DM, then the rectangles CM and DF are equal

The rectangles CM and AL are equal (I.36)

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For a proof showing that DM and LG are squares, see II.4

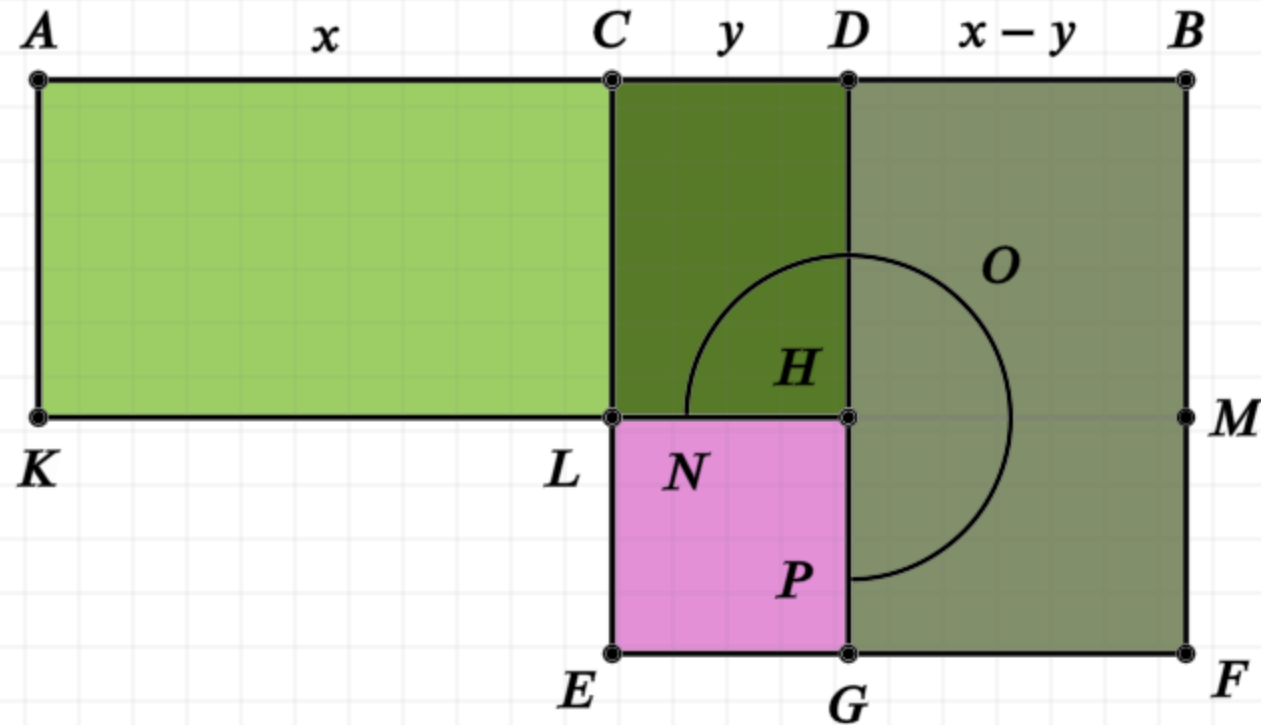
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LG is equal to the square on CD, add it to both AH and NOP, retaining the equality

But CF is equal to the square on CB, which is also equal to the gnomon NOP added to the rectangle LG, thus we have demonstrated the proof for this proposition

Proposition 5 of Book 2

If a straight line be cut into equal and unequal segments, the rectangle contained by the unequal segments of the whole together with the square on the straight line between the points of section is equal to the square on the half



$$AC = CB, \quad AD = AC + CD, \quad DB = CB - CD$$

$$CH = HF$$

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$$AH = NOP$$

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$$AD \cdot DB + CD^2 = NOP + LG = CF$$

$$AD \cdot DB + CD^2 = CB^2$$

In other words:

Let AB be a straight line, bisected at point C, and cut at an arbitrary point D

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Proof:

The complements CH and HF are equal (I.43), and if we add the rectangle DM, then the rectangles CM and DF are equal

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For a proof showing that DM and LG are squares, see II.4

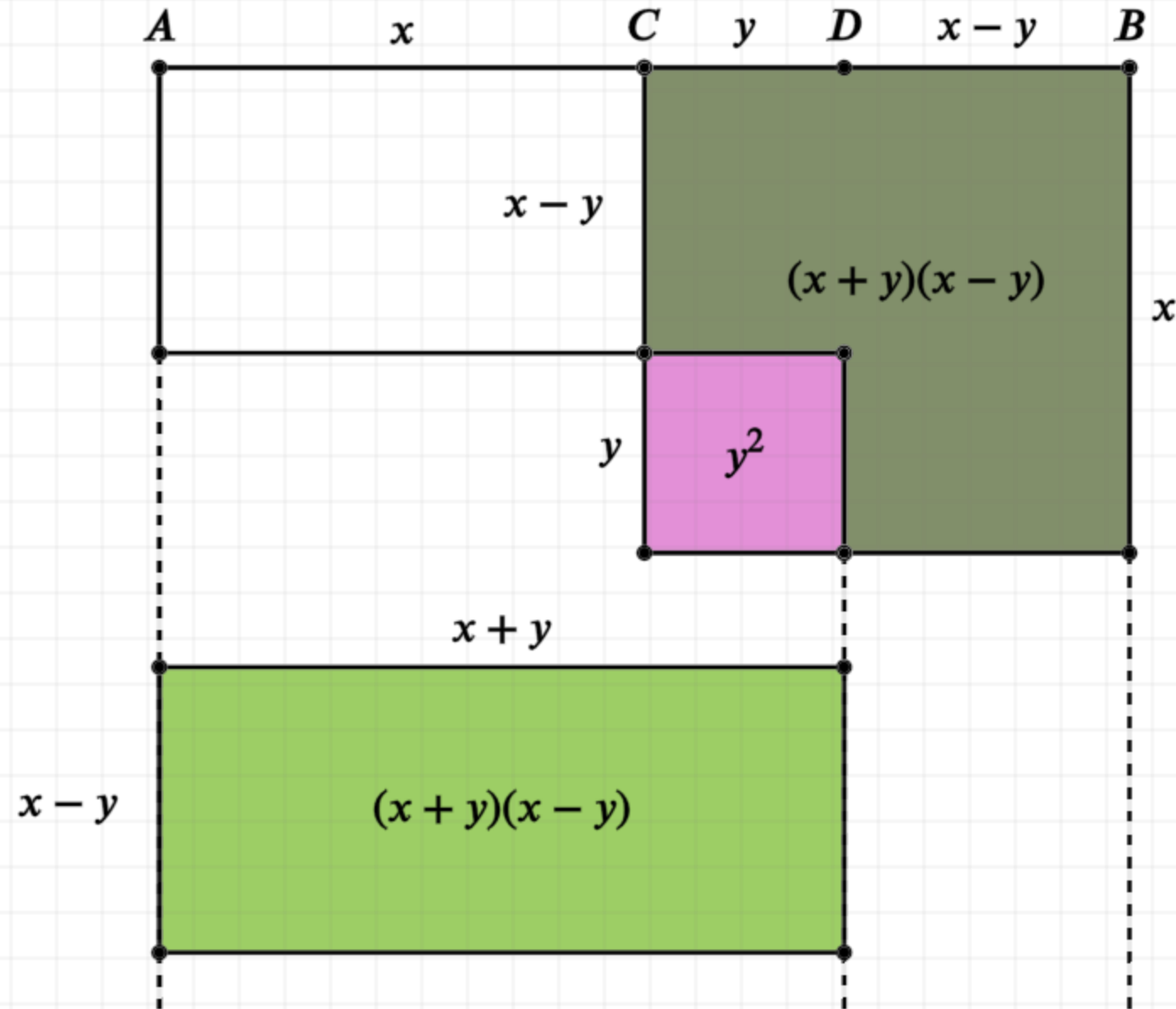
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Proposition 5 of Book 2

If a straight line be cut into equal and unequal segments, the rectangle contained by the unequal segments of the whole together with the square on the straight line between the points of section is equal to the square on the half



$$AD \cdot DB + CD^2 = CB^2$$

$$(x + y)(x - y) + y^2 = x^2$$

In other words:

Let AB be a straight line, bisected at point C, and cut at an arbitrary point D

$$AD \cdot DB + CD^2 = CB^2$$

Proof:

The complements CH and HF are equal (I.43), and if we add the rectangle DM, then the rectangles CM and DF are equal

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LG is equal to the square on CD, add it to both AH and NOP, retaining the equality

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